

RAHUL ARRA

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SUMMARY

Information Technology undergraduate at CBIT Hyderabad (CGPA: 9.03, May 2027) with hands-on experience building AI-integrated systems — LLM-based threat analysis, RAG pipelines, and Generative AI. Deployed scalable MERN applications on AWS with auto-scaling and load balancing. Passionate about building intelligent, cloud-native systems at scale.

EDUCATION

Undergraduate: B.E., Information Technology CGPA: 9.03 Graduating May 2027

Chaitanya Bharathi Institute of Technology, Hyderabad

Intermediate: CBSE, M.P.C (Mathematics, Physics, Chemistry) Percentage: 85 March 2023

Jawahar Navodaya Vidyalaya, Karimnagar

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, SQL

Web Development: React, Node.js, Express.js, REST APIs, HTML, CSS, WebSockets

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, ASG, CloudFront, ALB), Docker, Google Cloud Platform, Linux

Databases: MySQL, MongoDB, SQLite, JDBC

AI/ML: Machine Learning, Deep Learning, Generative AI, LLM Integration, Prompt Engineering, Gemini API

Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS

Big Data: Spark, HDFS, Kafka

Developer Tools: Git, Postman, VS Code

PROFESSIONAL EXPERIENCE

Vault Of Codes, Full Stack Intern June 2025 – Aug 2025

- Designed and implemented **10+ RESTful backend APIs** using Node.js and Express, contributing to core business logic and ensuring reliable data exchange across application workflows.
- Integrated backend services with frontend components, implementing input validation and structured error handling to deliver consistent and reliable API responses.
- Worked with MongoDB to design 5+ collections and optimize data structures, improving data integrity and query efficiency.
- Utilized Git for version control and Postman for API testing and validation during development.

ACADEMIC PROJECTS

AI-Integrated Honeypot with Chaos Engineering Simulation | AWS EC2 | Python | Docker | SQLite

- Architected a **fake SSH honeypot** on **AWS EC2** that captured and logged **500+ attacker sessions**, enabling real-time behavioral analysis.
- Integrated **LLM-based log analysis** (Groq API) to auto-classify attacker commands into **10+ threat categories** with **92% classification accuracy**.
- Engineered a chaos engine simulating latency injection, service disruption, and many other failure scenarios — reducing MTTD by **35%**.
- Containerized the application and attack simulation environment using **Docker**, enabling isolated, reproducible deployments across **10+ simulated attack scenarios**.

Real-Time Big Data Streaming & Analytics | Apache Flink | Kafka | Node.js | React

- Architected an event-driven pipeline using **Apache Flink** and **Kafka** to process **5,000+ events/hr** with **1s** end-to-end latency.
- Embedded **TextBlob NLP** into the streaming layer for real-time sentiment scoring and routing across **4 distinct categories**.
- Engineered a Node.js and Socket.io middleware bridge to broadcast a rolling window of 200+ live analytics to a React.js dashboard, eliminating **100%** of HTTP polling overhead.

Digital Health Portfolio (AWS Deployed) | MERN | AWS | RAG | Gemini API

- Deployed full-stack **MERN** application on **AWS EC2** with **S3** for secure, scalable storage of medical records, prescriptions, and health history.
- Scaled system to handle **300+ concurrent user requests** using AWS Auto Scaling Groups and Application Load Balancer with **99.9% uptime**.
- Built a **RAG pipeline** using vector search to retrieve health records and generate personalized suggestions, medication reminders, and insights via **Gemini API** — matching **10+ patient specific conditions** per query.

Online Movie Ticket Booking System | Java | JDBC | JSP | MySQL

- Developed a multi-tier application using **Java, JDBC, JSP, and MySQL** following OOP and MVC design principles, supporting **3 distinct user roles** (User, Admin, Theatre Manager).
- Enforced **ACID-compliant transactions** with pessimistic locking to eliminate concurrent booking conflicts.
- Implemented **Role-Based Access Control (RBAC)** with 3 privilege tiers for secure user and administrator operations.

CERTIFICATIONS & ACHIEVEMENTS

- **Google Cloud Computing Fundamentals** — NPTEL, 2024
- **Salesforce Certified AI Associate** — Trailhead, 2025
- **MongoDB Associate Developer Certification**, 2025
- **Oracle Cloud Infrastructure Certified AI Foundations Associate** — Oracle, 2025
- **Vault of Codes** — Backend Development Program, 2025
- Solved **350+ DSA problems** across LeetCode and GeeksForGeeks